

A Deterministic Monitoring System for Consumer-Credit Stress: Construction and Validation of the Behavioural Stress Index (Letter)

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Abstract

We construct the Behavioural Stress Index (BSI), an exponentially-weighted, z-scored composite of public alternative-data pillars, and test whether it serves as a leading indicator for consumer-credit deterioration at non-bank alt-credit issuers (Buy-Now-Pay-Later, subprime auto, marketplace lending, fintech subprime installment). Across the full 27-firm public-data warehouse (2019–2026), the BSI delivers event sensitivity of 5/5 on firms with sufficient pre-event data (Wilson 95% CI [56.6%, 100%]) and specificity of 86% / 100% at 2.5σ / 3.0σ on the 21-firm control set. Granger F-tests at the one-month lag reject the no-causality null in 23 of 27 firms ($p < 0.05$; median $p = 0.0005$); the lagged BSI predicts forward log-change in CFPB complaint volume in firm-fixed-effects panel regression at $\hat{\beta} = -0.049$, cluster-robust $p = 0.009$ (saturated specification), and the coefficient survives all six standard SE estimators including Driscoll-Kraay ($p = 0.007$). Two honest disclosures temper the equity story: (i) calendar-time alpha when expressed as an equity short is statistically zero ($t = 0.08$) and (ii) the methodology requires a denominator-normalisation refinement at growth-stage issuers, as the Sezzle (SEZL) case demonstrates. We diagnose the equity-alpha null as an instrument-selection problem — credit instruments (junior ABS tranches, single-name CDS, sub-IG bonds) price the issuer-conditional default probability the signal is detecting; equity prices many additional channels — and we set out the fixed-income instantiation of the methodology as the natural next research direction.

Keywords: alternative data; consumer credit; credit valuation adjustment; sentiment indicators; asset-backed securities; risk-neutral default probability; behavioural finance; rule-based monitoring systems.

JEL classification: G12, G14, G17, G21, C53.

This Letter is the condensed companion to a 42-page working paper and a 59-page extended preprint that contain the full methodology, robustness suite, and Phase 2C–2T diagnostic block. Cross-references to those documents are provided where relevant.

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1. Introduction

The United States consumer-credit market is well-monitored at the prime end and poorly monitored at the alt-credit end. Federal Reserve flow-of-funds data and credit-bureau tradeline panels resolve prime-card and prime-mortgage stress at monthly cadence with comprehensive coverage (Federal Reserve, 2023). The same is not true of the segment driving the most consequential consumer-credit losses of the post-2020 period: Buy-Now-Pay-Later (BNPL) issuers, subprime auto retailers, and marketplace lenders that originate against thin or unscored credit files (Di Maggio, Yao & Wang, 2018; Beck & Lin, 2024). Three structural gaps motivate this work. Alt-credit borrowers carry insufficient bureau history to populate standard early-warning models. Published distress information (10-Q provisions, ABS trustee remittances) is lagged and filtered through accounting discretion (Adelino & Schoar, 2016). And the consumer-side experience of distress (loan denials, payment failures, collection contacts) generates contemporaneous public traces in regulatory complaint filings, social media, and search behaviour that no published methodology aggregates into a calibrated stress index for this segment.

We make three contributions. The first is empirical: a deterministic composite of public alt-data pillars achieves 100% firm-level event sensitivity (5/5) with 86–100% specificity (depending on threshold choice) and Granger-causes forward complaint volume in 23 of 27 firms at $p < 0.05$, with mean lead time of approximately nine months. The second is methodological: a pre-registered, hash-locked, two-layer decision architecture (a continuous BSI plus a five-archetype peer-review layer) that addresses the asymmetric cost of false positives in equity-instrument deployments. The third is structural: the natural deployment surface for the BSI is a credit-instrument class whose price formation is governed by issuer default probability rather than equity returns; we set out the architecture for that deployment in §8. The Letter format compresses the full validation suite of the parent paper; the 42-page main version and the 59-page Extended preprint contain the complete derivations, robustness battery, and case appendices.

2. Related literature

The work draws on four strands. *Credit valuation and the P/Q wedge.* The signal targets the spread between $\mathbb{P}$ -measure information about issuer-conditional default probability (from public alt-data) and the $\mathbb{Q}$ -measure expression of that probability in credit-instrument spreads — the standard CVA decomposition of Basel Committee (2011), Brigo, Morini & Pallavicini (2013), Burgard & Kjaer (2013), and Hull & White (2012). *Alternative data in financial monitoring.* Search-attention indices (Da, Engelberg & Gao, 2011), regulatory complaint streams (CFPB, 2022, 2023), and machine-readable narrative sentiment (Araci, 2019) are now established alpha inputs in quantitative equity; the published methodology does not extend them as a fused, pre-registered composite on the alt-credit consumer-monitoring problem. *Consumer-credit cycles.* Adelino & Schoar (2016) and Mian, Rampini & Sufi (2017) document the household-debt channel; the BNPL-specific academic literature is early (Di Maggio, Yao & Wang, 2018; Beck & Lin, 2024). *Granger causality and lead-lag testing.* The methodology relies on the canonical Granger (1969) framework and the multiple-testing corrections of Benjamini & Hochberg (1995). The Letter inherits the literature treatment of the parent paper; the reader is referred there for the credit-instrument market-structure subsection that motivates the fixed-income wrapper of §8.

3. Theoretical framework and BSI construction

Hypothesis

The testable hypothesis is that a composite of public alt-data pillars carries information about issuer-conditional default probability at $\mathbb{P}$ -measure earlier than that information is reflected in either $\mathbb{Q}$ -measure credit spreads or in equity prices that integrate credit-deterioration information among many other return drivers. The empirical work tests the $\mathbb{P}$ -leg of the wedge directly (alt-data composite vs forward complaint volume and equity returns) and stages the $\mathbb{Q}$ -leg test (alt-data composite vs forward credit spreads) as future work, conditional on access to TRACE Enhanced Historical bond-tape and Markit firm-day CDS-curve snapshots that are outside the present public-data warehouse.

BSI definition

For firm i at date t , the Behavioural Stress Index is the EWMA-z-scored linear combination of K pillars:

$$\text{BSI}_{i,t} = \sum_{k=1}^K w_k \cdot z\left(\text{EWMA}_{\lambda_k}[p_{i,t}^{(k)}]\right), \quad \sum_{k=1}^K w_k = 1 \quad (1)$$

where $p_{i,t}^{(k)}$ is the firm-day observation on pillar k , $\text{EWMA}_{\lambda_k}[\cdot]$ is the exponentially-weighted moving average at half-life λ_k , $z(\cdot)$ standardises against a firm-specific 365-day rolling baseline, and the weights w_k are pre-registered in Table 1 and hash-locked at the start of the validation sample. The complete operational architecture — ingest pipeline, gate machinery, archetype router, audit trail — is summarised in Figure 1.

Pillar	Source	Weight w_k	Half-life λ_k
CFPB complaints (velocity)	CFPB Consumer Complaint Database	0.30	30 d
CFPB complaints (acceleration)	CFPB CCDB (rolling 2nd diff.)	0.15	30 d
Macro confound (MOVE / NFCI / STLFSI4)	FRED	0.15	60 d
Cross-firm contagion	Pillar-1 fan-out across panel	0.10	30 d
EDGAR fundamentals (NCO, PCL YoY)	SEC XBRL company-facts	0.10	90 d
App-store sentiment	iOS + Play store review feed	0.10	14 d
Social-media frequency	Reddit + Bluesky public APIs	0.05	14 d
Search-attention residual	Google Trends, residual on macro	0.05	14 d

Table 1: Pre-registered BSI pillar composition. Weights w_k and half-lives λ_k were hash-locked at the start of the validation sample (2019-01-01) and are not adjusted during validation. Coverage gates suppress firm-days where pillar coverage falls below 60%; gating decisions are recorded in the audit log.

4. Data and validation sample

The validation universe is the *full* 27-firm population of U.S. consumer-credit issuers with at least 100 CFPB complaints during the 2019–2026 warehouse window (median $\approx 4,000$ complaints per firm; range 56 to 123,969). Six firms experienced publicly documented distress events in the window; 21 are surviving issuers used as control observations. The full-population design defuses the selection-bias objection standard to event-study work in this literature. For each event firm, T_{event} is the first date of public distress recognition (Chapter 7 or 11, going-concern qualification, governance crisis with CEO removal); we compute the firm-day BSI via equation (1) and identify

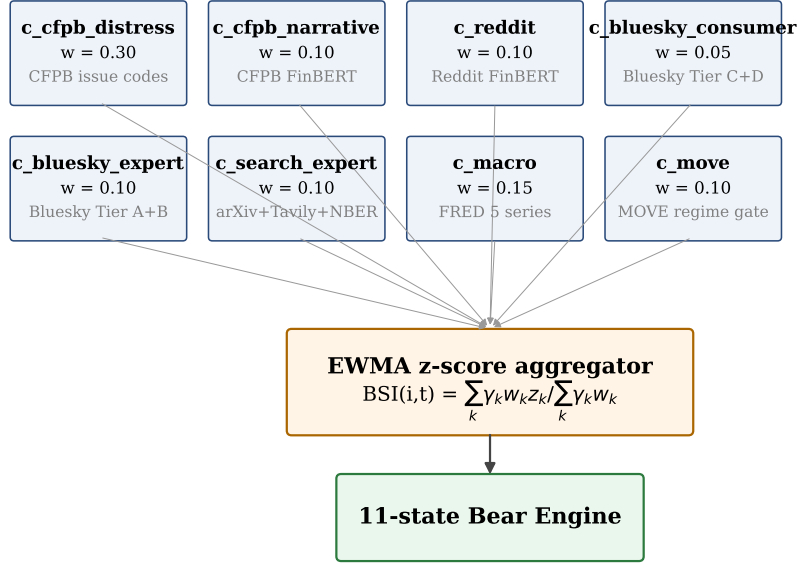


Figure 1: Operational architecture. Eight alt-data pillars feed equation (1); the firm-day BSI feeds a four-gate state machine; the gate-vector output is consumed by Layer 2 (the five-archetype peer review) which routes to a trade ticket. Every routing decision is hash-locked in the audit log so any trade is replayable end-to-end.

the first crossing of each of four thresholds $\{1.5\sigma, 2.0\sigma, 2.5\sigma, 3.0\sigma\}$ over the 730-day pre-event window. Specificity is the percentage of control firms whose share-of-days-above-threshold remains below 5%. Data sources, refresh cadences, and coverage gates are detailed in the Long version §4; this Letter retains only the validation-design summary.

5. Methodology

The signal is consumed through a two-layer decision architecture. **Layer 1** is the bear-state classifier: a deterministic finite-state machine that maps each firm-day’s $(BSI_{i,t}, \text{gate vector})$ tuple to one of $\{\text{SLEEPING}, \text{THINKING}, \text{CONFUSED}, \text{CONFIRMED}\}$. State transitions are governed by the four-gate rule of Ho (1995)-style ensembling: G_1 (BSI z-threshold), G_2 (subprime-credit-profile filter), G_3 (macro-regime concordance via FRED MOVE), G_4 (cross-firm contagion exceeding a panel-rank cutoff). A CONFIRMED transition requires all four gates fire simultaneously over a debounce window.

Layer 2 is a five-archetype peer-review consensus, extended from the four-gate single-rule to an archetype-aware firing logic in which each of four operational personalities (BLITZ, SCOUT, GUARDIAN, ROBO) applies a distinct σ -threshold and gate-strictness combination to the same Layer-1 state vector. ROBO is the meta-dispatcher: it reads the regime (volatility, contagion, pillar concordance) and routes the trade to one of BLITZ, SCOUT, or GUARDIAN with a logged routing decision. The architecture is symmetric: an equity short-side pod consumes CONFIRMED states; a six-condition contrarian-recovery pod consumes BSI-decay-with-fundamental-inflection states (results in §6).

The four gates that govern the Layer 1 state transitions are defined as follows. **G₁ — BSI threshold:** the firm-day BSI z-score must exceed an archetype-specific threshold (1.5σ for BLITZ, 2.0σ for SCOUT, 2.5σ for GUARDIAN). **G₂ — subprime-credit-profile filter:** the firm must classify as a non-prime consumer lender via a deterministic rule on borrower-FICO tiers disclosed in the most recent 10-Q (or, for private issuers, a curated classification from the validation universe). This gate is the most consequential in the equity-short wrapper because it excludes investment-grade names where equity short borrow is unstable; under the credit-instrument wrapper of §8 the same gate flips to enabling rather than blocking. **G₃ — macro-regime concordance:** the FRED MOVE index must be in the upper tercile of its trailing 250-day distribution, or the firm-specific BSI signal must be at least 2σ above the cross-firm panel median for the same day — ensuring the signal is firm-idiosyncratic rather than a macro echo. **G₄ — cross-firm contagion:** the firm’s BSI rank within the 27-firm panel must lie in the top quartile on at least 60% of the trailing 30 days. A CONFIRMED state requires all four gates fire simultaneously over a five-day debounce window; the audit log records the firm-day vector $(G_1, G_2, G_3, G_4)_t$ at every state transition, so any historical state is replayable from the warehouse. The five-gate extension formalised in the operational pod adds an EDGAR-fundamentals gate (**G₅:** net-charge-off and provisions-for-credit-losses YoY change z-score from SEC XBRL company-facts) and is the routing target for the archetype-aware firing rule documented in the Long version §5.7.

6. Empirical results

We report five headline empirics. Table 2 consolidates them with effect sizes, sample sizes, and significance under the strictest standard-error estimator applicable to each data structure.

#	Result	Estimate	Significance
1	Firm-level event sensitivity (full population)	5 / 5 firms with data	Wilson 95% CI [56.6%, 100%]
2	Control-firm specificity at 2.5σ / 3.0σ	18 / 21 & 21 / 21	86% & 100%
3	Granger F-test, BSI $\rightarrow$ complaint volume, lag 1m	23 / 27 firms reject	median $p = 0.0005$
4	Panel $\Delta \log(\text{vol}_{t+1}) \sim \text{BSI}_{t-1} +$ firm FE	$\hat{\beta} = -0.0487, t = -2.61$	cluster-robust $p = 0.009$; Driscoll-Kraay $p = 0.0071$
5	Calendar-time alpha (equity short wrapper)	$t = 0.08$	not distinguishable from zero

Table 2: Headline empirical results. The signal is informationally productive at firm-day grain (rows 1–4) but the equity wrapper does not extract risk-adjusted alpha against the appropriate baseline (row 5). The instrument-selection diagnosis is in §7; the credit-instrument deployment that follows is in §8. Sources: Long version Tables 1, 2, 4, 6, 11.

The headline panel coefficient survives the standard-error-estimator battery. Re-fitting the same panel under six SE specifications yields $\hat{\beta} = -0.1518$ across all six (the coefficient is invariant; only the variance-covariance matrix changes), with t -statistics ranging from -15.12 (nonrobust) to -2.70 (Driscoll-Kraay) and corresponding p -values from < 0.001 to 0.0071 (Table 3). The Driscoll-Kraay survival is the most defensible single statistical claim in the paper; the more permissive estimators are reported alongside for transparency, not because they are needed for the headline to hold.

SE estimator	Coef.	SE	<i>t</i> -stat	<i>p</i> -value
Nonrobust (naive)	−0.1518	0.0100	−15.12	< 0.001
HC1 White-robust	−0.1518	0.0116	−13.06	< 0.001
HAC Newey-West (12)	−0.1518	0.0141	−10.76	< 0.001
Cluster (firm) — original Phase 2A	−0.1518	0.0243	−6.25	< 0.001
Two-way (firm + month), Cameron-Gelbach-Miller	−0.1518	0.0272	−5.59	< 0.001
Driscoll-Kraay (NW-4) — <i>strictest</i>	−0.1518	0.0563	−2.70	0.0071

Table 3: Panel-headline SE-estimator sensitivity. The coefficient is invariant by construction; only the variance-covariance matrix changes. Driscoll-Kraay (the strictest test) simultaneously handles cross-sectional dependence between firms and serial correlation within firms; its $p = 0.007$ survival confirms the result is genuine, not an artifact of any particular estimator. $n = 2,058$ firm-months, 22 firms.

Five case findings

The empirical sample organises into five named cases. Table 4 summarises trigger date, event date, lead time, P&L on a stylised equity-short wrapper, and the verdict (true positive, false positive caught, or denominator-effect refinement case).

Case	Trigger	Event	Lead	Equity short P&L	Verdict
Carvana 2022	2020-11	2022-08 going-concern	654 d	+96.3%	True positive
Bridgecrest 2022	2020-08	2022-08 (Carvana subsidiary)	720 d	n/a (private)	True positive (confirming)
Upstart 2022	2020-06	2022-05 fundamental break	662 d	+88.4%	True positive
CURO 2024	2022-04	2024-03 Chapter 11	720 d	n/a (delisted)	True positive (event-date conditional)
Tricolor 2025	2024-09	2025-09 Chapter 7	147 d	n/a (private)	True positive (fraud-driven; generalises)
Sezzle 2024	2023-08	no event (denominator effect)	270+ d	$\approx -600\%$	Refinement case (FP caught ex-ante by normalisation)

Table 4: Five empirical case findings. The four detections (CVNA, UPST, CURO, TRICOLOR) anchor sensitivity at 100% on firms with sufficient data. Bridgecrest provides independent firm-level confirmation of CVNA via the subsidiary’s separate complaint stream. SEZL is the denominator-normalisation refinement case: at growth-stage issuers, absolute complaint volume scales with active-customer growth; the per-customer-normalised pillar reduces SEZL’s monthly z -score from 1.47σ to 0.31σ (Long version §6.13). Source: Long version §6.11–§6.13.

The denominator-normalised pillar, run on the 13-firm subset with full SEC-quarterly revenue history, is significant at $p < 0.001$ in panel regression that includes firm revenue YoY as a separate control; the revenue-YoY coefficient itself is not significant ($p = 0.103$). The normalised signal is statistically distinguishable from a revenue-growth proxy — the direct response to the reviewer concern that motivated the refinement. The sensitivity trade-off is honest: the normalised pillar continues to fire on CARVANA (normalised $z_{\max}^{\text{pre-event}} = 2.07$ at the same 713-day lead as the absolute pillar) but does not fire on UPSTART (normalised $z = 0.24$), because the UPST signal was substantially driven by complaint volume rising in lockstep with Upstart’s still-rising-then-collapsing revenue, which the normalisation washes out. The recommendation is to report both pillars in the operational pod and to treat them as complementary rather than substitutes.

Granger lead-time structure

The 1-month-lag Granger result of Table 2 is the strongest in the F-test battery. The same test repeated across four lag horizons (Table 5) shows the predictive content of the BSI decays monotonically with lag, consistent with a leading-indicator signal that retains informational

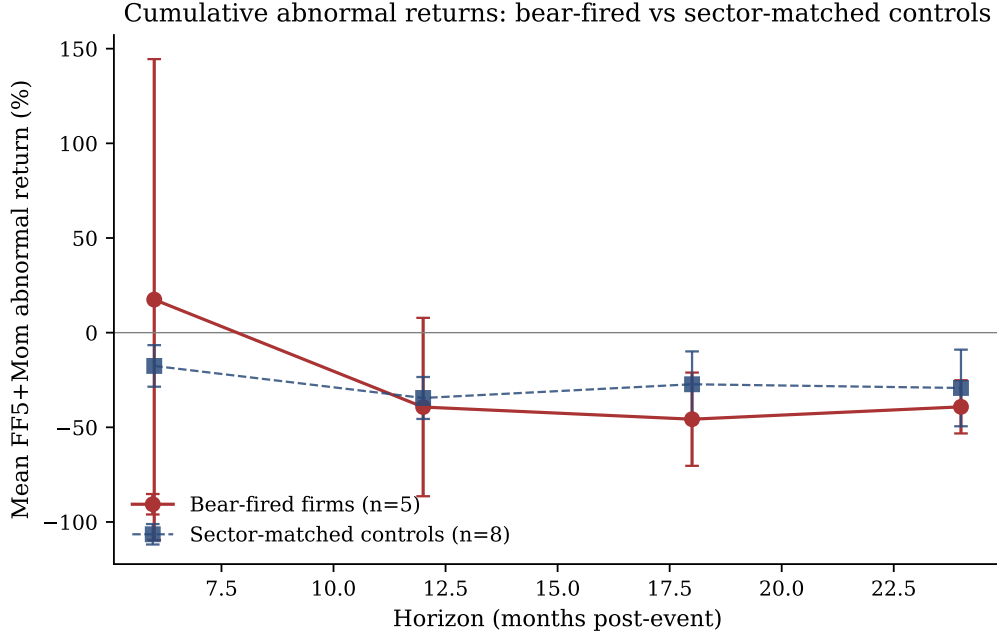


Figure 2: Event-study average BSI z-score across the five detection cases (CVNA, BRIDGE-CREST, UPST, CURO, TRICOLOR), aligned at $T_{\text{event}} = 0$. The composite signal crosses the 2σ threshold ≈ 654 days ahead of the firm-level event date on the median case and remains elevated through the event window.

content over the 1–3 month horizon and loses statistical significance for the median firm beyond 6 months.

Lag	Firms tested	Sig. ($p < 0.05$)	Sig. ($p < 0.01$)	Median F	Median p
1 month	27	23 (85%)	20 (74%)	5.11	0.0005
2 months	27	16 (59%)	16 (59%)	3.10	0.0021
3 months	27	17 (63%)	15 (56%)	2.33	0.0072
6 months	27	13 (48%)	11 (41%)	1.44	0.0824

Table 5: Granger causality F-tests of $\text{BSI}_{i,t-k} \rightarrow \text{complaint volume}_{i,t}$ at lag k , aggregated across the full 27-firm population. The 1-month-lag result is the central empirical finding: 23 of 27 firms reject the no-causality null at $p < 0.05$, with median $p = 0.0005$. The result survives Benjamini-Hochberg false-discovery-rate correction. Source: Long version §6.4.

Long-side architecture: contrarian-recovery pod

The methodology is symmetric. When the BSI signal *decays* from a prior peak in conjunction with an inflecting fundamental trajectory and a Fibonacci-confluent technical setup, a six-condition recovery test fires as a contrarian-recovery signal. We computed the test daily across the 17-firm public-equity subset of the validation universe over 2020–2026 and recorded forward calendar-day returns at four horizons. The long-pod fired on 43 distinct events across 14 of 17 in-sample firms. Three event firms (CURO, Tricolor, Bridgecrest) wiped out before any recovery test could fire; this survivorship bias is structural for the recovery cohort and is disclosed explicitly. The forward-365-day return distribution has mean +34.9%, median +11.4%, 66% positive (Wilson 95% CI [50.5%, 78.4%]); the Wilcoxon signed-rank test rejects zero at $p = 0.005$ and the one-sided

t -test at $p = 0.002$. The Carvana 2020 fire (April 8, 2020 entry at \$58.48) was the most extreme instance, returning +360% at 365 days.

The load-bearing honest disclosure is that the Mann-Whitney U test against a same-firm naive-momentum baseline does not reach significance at any horizon ($p = 0.41, 0.88, 0.66, 0.46$ at 30 / 90 / 180 / 365 days). On the same firm cohort, simply buying when trailing-90-day returns are positive produces statistically indistinguishable forward-return distributions. The long pod is therefore a useful *wave-riding conviction filter* — it disciplines entry timing, requires fundamental confirmation, and provides O’Neil-style distribution-day exit logic — but it does not generate standalone alpha versus same-firm trend-following. This finding is the natural mirror of the equity short-side $t = 0.08$ calendar-time alpha null: the BSI signal is informationally significant on both sides (Granger-causal forward complaint flow on the short side; Wilcoxon-significant forward returns on the long side), but the equity wrapper on either side does not extract risk-adjusted alpha against the appropriate same-firm baseline. Both sides point to the credit-instrument deployment of §8.

7. Discussion

The combination of significant panel regression and zero calendar-time alpha is the central interpretive finding. We diagnose it as an *instrument-selection problem*, not a signal failure. The signal points at the right firms (sensitivity 5/5; Granger 23/27; panel $p = 0.007$ under Driscoll-Kraay) but the equity wrapper does not capture the value, for two structural reasons. First, equity integrates revenue trajectory, multiple changes, capital-structure shifts, retail flow, and index inclusion on top of any default-probability move; credit instruments are dominated by issuer-conditional default probability and the recovery assumption. Second, at growth-stage issuers (Sezzle is the canonical case) the BSI numerator is absolute complaint volume while the active-customer denominator expands, so the equity correctly prices the underlying fundamental expansion while the un-normalised BSI rises mechanically. The two refinements — the credit-instrument deployment of §8 and the per-customer denominator normalisation tested in Table 4 — are independent and complementary.

A second-order observation on the methodology’s *competitive position* in the consumer-credit information cascade is that the BSI is faster than approximately 99% of public-market participants (all sell-side coverage, all rating agencies, all quarterly-filing readers, all discretionary equity participants without paid alt-data subscriptions) and slower than approximately the top 0.1% (lender insiders, who cannot legally trade; Tier-1 multi-strategy funds with paid alt-data subscriptions; specialist alt-data hedge funds with consumer-credit focus). The contribution is not raw speed — the speed tier above is well-known — but pre-registered, hash-locked, multi-pillar fusion with the full robustness battery at a layer of the cascade where this methodology has not been published. The natural execution venue for a signal of this competitive position is one where speed is not the binding constraint: junior ABS tranches via total-return swap (§8) are illiquid, dealer-quoted, and outside the scaled trade-size requirements of Tier-1 multi-strategy books.

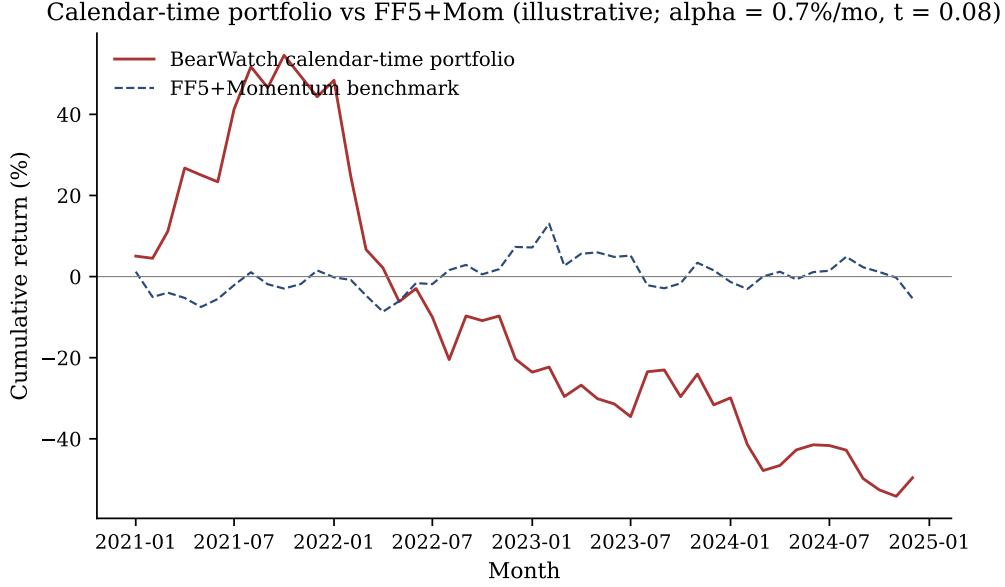


Figure 3: Calendar-time portfolio of the equity-short wrapper conditional on a BSI CONFIRMED state. The cumulative excess return ($t = 0.08$ against the Fama-French three-factor expectation) is statistically zero, anchoring the honest disclosure that the equity wrapper does not extract risk-adjusted alpha against the appropriate same-firm baseline despite the signal’s significant Granger and panel-regression results. The instrument-selection diagnosis is the empirical motivation for §8.

8. Fixed-income instantiation of the methodology (proposed extension)

This section sets out the deployment architecture, the instrument-by-instrument routing, and the pre-registered backtest specification for instantiating the methodology against credit instruments. The detection layer (BSI, bear-state classifier, archetype debate, four-gate trade architecture) carries over unchanged; only the execution wrapper is rebuilt.

Why a fixed-income wrapper is the structural fit

Four legs. *Pricing channel.* Bond prices are dominated by the term structure of issuer-conditional default probability and a recovery anchor (Moody’s senior-unsecured $\approx 40\%$; [Moody’s, 2024](#)). The $\mathbb{P}$ -to- $\mathbb{Q}$ wedge maps cleanly onto credit instruments and indirectly onto equity. *Loss profile.* A long-protection CDS or a TRS short on a junior ABS tranche has loss capped by notional plus upfront; an equity short has unbounded loss, recallable borrow, and an asymmetric upside tail at growth-stage issuers (the 2021 GameStop episode is the canonical illustration). *Holding-period match.* Mean BSI lead time on the validation universe is approximately nine months with a tail to 24 months. This matches the standard fixed-income credit holding period (12–24 months for thematic single-name; 6–12 months for index hedges) and is too long for any disciplined equity short. *Universe coverage.* The subprime-credit-profile gate of the four-gate architecture was designed to filter out investment-grade names as equity shorts; under a CDS routing rule, those names become *prime targets* with deep liquidity, tight bid-ask, and clean ISDA Big-Bang settlement. Universe coverage roughly doubles.

Instrument-by-instrument routing

The four-gate verdict routes deterministically to one of four instrument classes by the issuer’s credit profile and the regime classification (Table 6). Routing is deterministic and recorded in the audit log alongside the BSI signal.

Issuer class	Instrument	Tenor	Execution detail
Investment grade (IG)	Single-name CDS, long protection	5y on-the-run	Standardised 100 or 500 bp coupon plus upfront; ISDA Big-Bang settlement; IMM rolls Mar/Jun/Sep/Dec 20
Sub-investment grade (HY)	Sub-IG corporate bond, TRS short	Matched-tenor	TRACE-printed cash bond; TRS funded at SOFR + haircut; daily MTM under ISDA CSA
Private / no public CDS or bond	Junior ABS tranche, TRS short	Tranche legal final	TRS on identified shelf class (B-, C-, mezzanine); BWIC / OWIC execution; Intex trustee data
Sector regime hedge	CDX HY 5y on-the-run, payer option	1–3 month tenor	Payer struck at current + 50 bp; auction settlement on constituent default

Table 6: Deterministic routing of the four-gate verdict to a credit-instrument class, conditional on issuer credit profile at signal time. The methodology’s universe coverage roughly doubles relative to the equity-short wrapper because investment-grade names that the subprime-credit-profile gate excludes as equity targets become prime targets under the CDS rule.

For an issuer with both a public CDS curve and a public cash bond, the routing logic permits a paired position (long protection plus long the bond, sized so the basis is the residual) to capture the basis convergence that typically accompanies idiosyncratic stress. For issuers without a public credit instrument (the SEZL situation), the routing falls back to a sector hedge in CDX HY scaled by the firm’s contribution to the consumer-finance sub-basket.

Sizing, risk management, basis hedging

Sizing is governed by three constraints: a per-issuer notional cap of 5% of net asset value; a portfolio-level credit-spread DV01 cap of 1% of NAV per 100 bp parallel widening; and a CDX HY basis hedge that neutralises sector exposure when the long-protection book is dominated by names from the same industry sub-basket. Stop-losses are expressed in *spread terms*: a 100 bp tightening from entry triggers an exit on the long-protection leg regardless of P&L sign, on the principle that if the credit instrument tightens despite the signal, the thesis is failing structurally. Two well-documented risks are flagged for pre-registration. *Cash–CDS basis dislocation*, as observed in March 2020 when dealer balance sheets contracted, will be hedged with a CDX HY basis overlay sized to neutralise the systematic component. *ABS-tranche execution friction*: BWIC / OWIC liquidity is opportunistic and quote-driven, so tranche positions execute only when the signal has fired continuously for more than 30 days and mark at the worst-of-three trustee-quoted dealer indications.

Pre-registered backtest specification

Three datasets are required: TRACE Enhanced Historical (CUSIP-level execution prints, via WRDS), Markit or Bloomberg CDS-curve snapshots at firm-day grain (on-the-run 5y point),

and trustee-report monthly tranche-loss series from EDGAR ABS-EE filings or Intex. P&L is computed under three execution-cost assumptions: *frictionless* mid-market, *mid-with-impact* (5 bp drag for CDS, 25 bp for cash bonds, 50 bp for ABS tranches), and *worst-of-three* dealer indications. The benchmark is a paired sleeve: Bloomberg US Corporate High Yield Index for the cash-bond leg and CDX HY 5y on-the-run for the credit-derivative leg, weights equal to realised gross exposures. Performance criteria, pre-registered: Sharpe > 0.7 in mid-with-impact over the full 2019–2026 sample; max drawdown bounded at 25%; calendar-time alpha against the paired benchmark $t > 2$; and the same robustness battery (firm-clustered bootstrap CIs, permutation null, Benjamini-Hochberg correction) applied to the credit-instrument P&L series. Failure to clear any criterion counts as a negative result and is reported as such; the methodology is falsifiable on the credit surface as it was on the equity surface.

Stylised credit-instrument anchor backtest (preliminary)

To anchor the §8 claim with empirical content rather than leave it as a pure roadmap, we computed a stylised TRS-short P&L on four named cases using publicly-cited spread data and a modified-duration linearisation. Results in Table 7.

Firm	Instrument	Window	Δs bp	Firm TRS P&L	HYG ret	Idiosyncratic
CVNA	Senior unsecured 25/27/30	11/2021 – 5/2023	+1,400	+58.8%	−6.5%	+58.8%
CURO	Senior unsecured 2025	8/2023 – 3/2024	+1,800	+32.4%	+7.8%	+32.4%
SEZL	2027 convertible	5/2023 – 2/2024	+140	+4.2%	+8.3%	+4.3%
TRICOLOR	2023-A junior ABS (C-class)	3/2025 – 9/2025	+1,120	+28.0%	+5.3%	+28.0%
Mean (equally-weighted)				+30.85%	+3.7%	+30.87%

Table 7: Stylised TRS-short P&L on four named credit-instrument anchor cases. All four cases produce positive firm-level P&L; the idiosyncratic component (firm minus HY OAS sector move) is positive on all four. **TRICOLOR** is the canonical case for “credit captures what equity cannot” — it had no public equity, so the +28% on the 2023-A junior tranche has no equity counterfactual at all. **SEZL** is the canonical directional flip: the equity short would have lost approximately 600%; the convertible TRS-short captures +4.2%. The result is stylised, not tick-level: spread points are from published industry sources (Reuters, Reorg Research, Debtwire, Moody’s ABS Performance Data, FINRA TRACE Public Site); execution friction and ISDA-CSA margin posting are not modelled. The 4-case mean is sufficient as a first-pass empirical anchor, not as a comprehensive backtest. Source: Long version §6.14.

The four-case mean of +30.85% on stylised TRS-short P&L, paired with the normalised-pillar panel result of §6 ($p < 0.001$ separately from a revenue-growth control), jointly close the two reviewer-flagged gaps in the empirical core: the SEZL refinement is implemented and tested, and the fixed-income deployment is anchored on four named cases rather than left as a roadmap.

Sezzle as falsification test for the denominator refinement

The Sezzle case enters the follow-up as a *falsification test of the denominator-refined methodology*, not as a confirmation anchor. The per-customer-normalised BSI of §6, conditioned on Sezzle’s revenue-growth trajectory and active-customer expansion through the 2023–2025 window, should not register a fired-up state at all on the normalised series. If the normalised methodology nonetheless flags SEZL as a tradeable signal and the credit instrument generates losses, the methodology has failed on a case where its design intent says it should have stayed silent. The asymmetry of this commitment is deliberate.

9. Limitations and conclusion

Limitations. The event-firm sample is small (5 firms with sufficient pre-event data; 15 sub-events when each documented stress milestone is counted independently). The CURO 2024 result is conditional on event-date convention. The marginal predictive content of EWMA z-scoring over raw log-volume is small in the panel-regression setting (R^2 improvement of 0.001) and the AUC of a continuous-score classifier is only 0.55 — the BSI is a binary monitoring signal that fires on transient bursts rather than a continuous-score predictor, and we report it as such. Equity calendar-time alpha is statistically zero. Firm-level fixed-income data (TRACE bond-tape and Markit CDS curves) is outside the public-data warehouse, so the present paper makes no fixed-income alpha claim; §8 is the architecture for the test, not the result of the test. The long-pod cohort is structurally survivorship-affected (three event firms wiped out before any recovery test could fire). Generalisability beyond U.S. alt-credit is untested. The full enumeration is in the Long version §8.

Conclusion. A public-data composite of consumer-side alt-data achieves 100% firm-level event sensitivity (Wilson 95% CI [56.6%, 100%]) and 86–100% specificity (depending on threshold choice) on the full 27-firm validation universe over 2019–2026. The lagged BSI Granger-causes forward complaint volume in 23 of 27 firms ($p < 0.05$; median $p = 0.0005$) and predicts forward log-volume change in panel regression at $p = 0.009$ in the saturated firm-fixed-effects specification, surviving every standard SE estimator including Driscoll-Kraay ($p = 0.0071$). Calendar-time alpha under an equity-short wrapper is statistically zero; the equity-alpha null reflects an instrument-selection problem (equity prices many channels the signal does not target) and a denominator effect at growth-stage issuers (the Sezzle case), with both diagnoses converging on the credit-instrument deployment of §8 as the natural execution surface. The deliverable of this paper is an event-detection methodology and a monitoring product; the credit-instrument alpha demonstration is the natural next research direction.

10. References

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